Exercise 8

Convolutional Neural Networks

1. Parameters – Weight sharing

We aim to analyze color images of size 100x100 pixels.

a) Suppose we design a fully connected multilayer perceptron (MLP) with **50 neurons** in the first hidden layer. How many **trainable parameters** does this layer contain?

b) Now consider a Convolutional Neural Network with five 3x3 filters (kernels). How many **trainable parameters** does the convolutional layer contain?

c) What is the difference between a **parameter** and a **hyperparameter**?

2. Data Preparation "Cats vs. Dogs"

Continue working with Jupyter Notebooks.

Please open the notebook `CNN_Data_Preparation.ipynb` and follow the instructions provided within! Answer the questions given in the end!

3. Jupyter Notebook – Convolutional Neural Networks

Continue working with Jupyter Notebooks.

Please open the notebook `CNN_Training.ipynb` and follow the instructions provided within.

Note:
Make sure that you've completed Exercise 2 successfully! Check that training **and** validation data is available in a directory `'data-dogs-vs-cats'` in the same directory as your notebook (your workspace), so it can be loaded correctly.